

power, respectively. In a similar manner the power required by each supercharger to compress a pound of air per second at other conditions of pressure difference and speed was determined and is presented in the following table.

TABLE II
HORSEPOWER REQUIRED BY EACH SUPERCHARGER TO
DELIVER ONE POUND OF AIR PER SECOND

Supercharger Speed - 1,000 r.p.m.

Pressure difference, in. of Hg	0	12	15
N.A.C.A. Roots supercharger, hp	0.294	42.6	64.5
Powerplus supercharger, hp	3.84	49.6	90.6

Supercharger Speed - 2,000 r.p.m.

Pressure difference, in. of Hg	0	12	15
N.A.C.A. Roots supercharger, hp	0.954	39.7	57.1
Powerplus supercharger, hp	6.57	39.55	59.8

Supercharger Speed - 2,500 r.p.m.

Pressure difference, in. of Hg	0	12	15
N.A.C.A. Roots supercharger, hp	1.342	39.5	57.6
Powerplus supercharger, hp	9.00	39.3	56.7

A study of Table II shows that below 12 inches of mercury pressure difference the results are more favorable to the Roots supercharger whereas from 12 to 15 inches of mercury pressure difference the performances are about equal at speeds of 2,000 revolutions per minute and above. At high pressure differences (above 15 inches of mercury) the performance of the Roots supercharger is superior to that of the Powerplus except at speeds of 2,500 revolutions per minute or higher where the performance of the latter is slightly better. As designed, this supercharger would be most satisfactory for conditions where the pres-